

Guilherme H. F. Azevedo

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EDUCATION

Universidade Federal de Minas Gerais
Doctor of Philosophy in Zoology

Belo Horizonte, Brazil
June 2016

“Phylogeny, taxonomy and macroevolution of copulatory structures in Gnaphosidae genera (Araneae: Gnaphosoidea).”

Universidade Federal de Minas Gerais
Master of Sciences in Ecology, Conservation and Wildlife Management

Belo Horizonte, Brazil
February 2012

“Systematics and evolution of the Brazilian wandering spiders (Ctenidae: *Phonentria*) based on molecular, morphological and ecological evidence.”

Universidade Federal de Minas Gerais
Bachelor of Science in Biology, emphasis in Zoology

Belo Horizonte, Brazil
August 2009

“Morphometrics of copulatory structures in three species of Brazilian wandering spiders (Ctenidae: *Phonentria*).”

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

San Diego State University

San Diego, CA
August 2019 – March 2023

- Prepare reduced representation genomic libraries (UCEs, ddRAD) for high-throughput sequencing.
- Prepare samples of spider tissue for acquiring RNAseq data.
- Bioinformatic processing of high-throughput sequencing data.
- Perform frequentist and Bayesian statistics for population genomics and phylogenetics.
- Adapt and develop bioinformatic workflows and analytical methods for population genomics and phylogenetics.
- Fieldwork to collect samples of spiders in CA, AZ, NV, UT, Baja California, Baja California Sur.
- Lead writing of peer-reviewed research papers.
- Train students in the preparation of reduced representation genomic libraries (UCEs, ddRAD).
- Communicate results in conferences.

Postdoctoral Researcher

Museo Argentino de Ciencias Naturales

Buenos Aires, Argentina
August 2017 – March 2023

- Prepare reduced representation genomic libraries (UCEs, ddRAD) for high-throughput sequencing.
- Perform PCR amplification of target genes for Sanger sequencing
- Bioinformatic processing of high-throughput sequencing and Sanger sequencing data.
- Perform Bayesian analyses for population genomics and phylogenetics.
- Fieldwork to collect samples of spiders in Argentina and Chile.
- Acquire morphological data for phylogenetic analyses.
- Lead writing of peer-reviewed research papers.
- Train students in PCR amplification of target genes for Sanger sequencing.
- Communicate results in conferences.

Postdoctoral Researcher

Universidade Federal de Minas Gerais

Belo Horizonte, Brazil
December 2016 – July 2017

- Prepare reduced representation genomic libraries (RAD) for high-throughput sequencing.
- Teach Invertebrate Zoology Undergrad Course.

Lecturer

Universidade Federal de Minas Gerais

Belo Horizonte, Brazil
August 2015 – August 2016

- Teach Invertebrate Zoology Undergraduate Course: Lab and lecture sections.

Lecturer

Universidade Federal de Minas Gerais

- Teach Invertebrate Zoology Undergrad Course: lab and lecture sections.
- Teach Arachnology Undergrad Course: lab and lecture sections.

Belo Horizonte, Brazil
March 2012 – December 2013

Undergraduate Research and Curatorial Assistant

Universidade Federal de Minas Gerais

Belo Horizonte, Brazil

August 2006 – July 2009

- Collect morphometric data of spider species.
- Perform multivariate statistical analyses of morphometric data.
- Perform standardized collecting protocols for structured ecological surveys.
- Sort and identify arachnids.
- Perform statistical analyses of diversity data for ecological survey reports.
- Lead writing of peer-reviewed research paper.
- Maintenance, organization, strategic growth and digitizing zoological research collection of invertebrates in ethanol.

Undergraduate Research Intern

Universidade Federal de Minas Gerais

Belo Horizonte, Brazil

August 2005 – December 2005

- Perform PCR amplification of target genes for Sanger sequencing.
- Lab organization and cleaning.

SKILLS

Laboratory: DNA isolation, library preparation for high-throughput sequencing, PCR, Sanger sequencing, gel electrophoresis, scanning electron microscopy, stereoscope microscopy, handling, preparation and curation of specimens and tissue sample for molecular and morphological use, proficient with multi-channel pipettes, 96 and 384-well plates.

Programming: Bash, R, Python.

Bioinformatics: processing of high-throughput sequencing data (QC, filtering, mapping, variant calling, orthology assessment, sequence alignment), population genomic analyses, maximum likelihood and Bayesian phylogenetics.

Applications & Packages: astral, bayestraits, bcftools, beast2, bioawk, bioconda, blast, busco, bwa, cd-hit, dendropy, dsuite, fastqc, feems, iqtree, mafft, mitofinder, orthofinder, phyluce, pilon, revbayes, samtools, seqtk, spades, stells2, structure, treeshrink, trimmomatic, vcftools.

Leadership & Teamwork: initiative, adaptability, problem solving, public speaking, and mentoring skills.

Languages: English, Portuguese, Spanish.

Other Selected Skills: Microsoft Office suite, Google Drive suite, version control with Git, documentation with markdown and Rmarkdown, comfortable working on a HPC environment.

PROFESSIONAL DEVELOPMENT HIGHLIGHTS

University of California Davis Bioinformatics Core RNA Sequencing Workshop

University of California, Davis

Davis, CA

August 2019

- Workshop centered around RNA-seq analysis, including experimental design, data generation, and post-sequencing processing and analyses.

SELECTED PUBLICATIONS (COMPLETE LIST AT [GOOGLE SCHOLAR](#))

Azevedo, G.H.F., Maddison, W. and Hedin, M. 2024. Phylogeny and biogeography of harmochirine jumping spiders (Araneae: Salticidae). *Molecular Phylogenetics and Evolution*, 197, p. 108109. <https://doi.org/10.1016/j.ympev.2024.108109>.

Azevedo, G.H.F., Blair, J. and Hedin, M. 2023. Evaluating possible anthropogenic impacts on gene flow and loss of genetic diversity in endangered Madla Cave Meshweaver spiders (Hahniidae, *Cicurina madla*). *Conservation Genetics*, 25, p. 149–164. <https://doi.org/10.1007/s10592-023-01561-y>.

Azevedo, G.H., Bougie, T., Carboni, M., Hedin, M. and Ramírez, M.J., 2022. Combining Genomic, Phenotypic and Sanger Sequencing Data to Elucidate the Phylogeny of the Two-Clawed Spiders (Dionycha). *Molecular Phylogenetics and Evolution*, 166, p. 107327. <https://doi.org/10.1016/j.ympev.2021.107327>